

# Cmod A7-35T Internal IP Peripheral Reference

**Platform:** Xilinx Artix-7 (xc7a35tcpg236-1) — Cmod A7-35T  
**Processor:** MicroBlaze RISC-V  
**System Clock:** 100 MHz (PLL from 12 MHz on-board oscillator via Clocking Wizard)  
**Toolchain:** Vivado & Vitis 2025.2

## 1. Processor Core & System Infrastructure

### 1.1 MicroBlaze RISC-V ( `microblaze_riscv_0` )

| Parameter              | Value                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| IP Version             | <code>xilinx.com:ip:microblaze_riscv:1.0</code>                                                                                   |
| Instruction / Data Bus | 32-bit LMB (Local Memory Bus) + AXI4 Data Port                                                                                    |
| Debug Interface        | Enabled ( <code>C_DEBUG_ENABLED = 1</code> )                                                                                      |
| Multiply / Divide      | Enabled ( <code>C_USE_MULDIV = 2</code> )                                                                                         |
| Bit Manipulation       | Bitmanip A/B/C/S all enabled                                                                                                      |
| Cache                  | 16 KB I-Cache + 16 KB D-Cache, write-through, 32 B lines, cacheable range <code>0x6000_0000 – 0x6007_FFFF</code> (exact SRAM fit) |

**Description:** The main processor core. Executes user firmware and accesses all peripherals through the AXI SmartConnect interconnect.

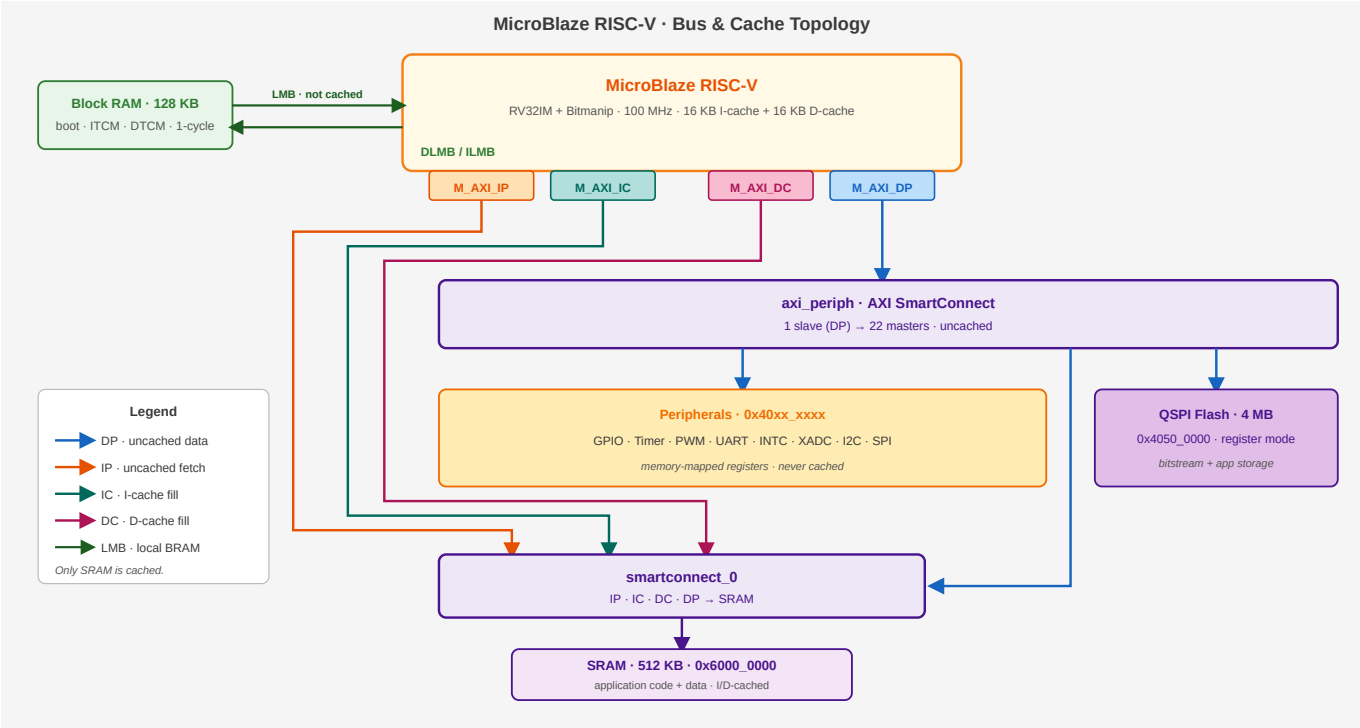
### 1.2 Local Memory

| Parameter              | Value                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------|
| IP                     | <code>lmb_v10</code> (LMB Bus) + <code>lmb_bram_if_cntlr</code> + <code>blk_mem_gen</code> |
| Data Bus (DLMB)        | <code>0x0000_0000 – 0x0001_FFFF</code> (128 KB)                                            |
| Instruction Bus (ILMB) | <code>0x0000_0000 – 0x0001_FFFF</code> (128 KB)                                            |
| Memory Type            | True Dual-Port Block RAM                                                                   |
| ECC                    | Disabled                                                                                   |

**Description:** Instruction and data local memory implemented with FPGA Block RAM — functionally this MCU's tightly-coupled memory (TCM): the ILMB/DLMB ports play the same role as ITCM/DTCM on other MCU cores (e.g. Cortex-M7), giving 1-cycle access outside the cache path. Layout: `0x0000 – 0x7FFF` (32 KB)

holds the UART bootloader (restored from flash at every configuration); `0x8000 – 0xFFFF` (32 KB) is the application's ITCM for interrupt handlers and timing-critical code ( `ITCM_FUNC` , copied out of the SRAM image by `tcm_init()` at startup); `0x10000 – 0x1FFFF` (64 KB) is the DTCM, holding the stack (top-down from `0x20000` ) and `DTCM_DATA` fast data.

1.3 AXI SmartConnect ( `microblaze_riscv_0_axi_periph` )



| Parameter    | Value                                       |
|--------------|---------------------------------------------|
| IP Version   | <code>xilinx.com:ip:smartconnect:1.0</code> |
| Master Ports | 22 (M00 – M21)                              |
| Slave Ports  | 1 (S00, connected to MicroBlaze M_AXI_DP)   |

**Description:** AXI interconnect crossbar that routes processor data transactions to 22 peripheral endpoints.

1.4 AXI Interrupt Controller ( `microblaze_riscv_0_axi_intc` )

| Parameter         | Value                                    |
|-------------------|------------------------------------------|
| IP Version        | <code>xilinx.com:ip:axi_intc:4.1</code>  |
| AXI Base Address  | <code>0x4040_0000</code>                 |
| Fast Interrupt    | Disabled ( <code>C_HAS_FAST = 0</code> ) |
| Interrupt Sources | 8 (merged via <code>ilconcat</code> )    |

Interrupt Mapping:

| Channel | Source | Description |
|---------|--------|-------------|
|---------|--------|-------------|

| Channel | Source   | Description                     |
|---------|----------|---------------------------------|
| In0     | timer_0  | System timer 0 interrupt        |
| In1     | timer_1  | System timer 1 interrupt        |
| In2     | timer_2  | System timer 2 interrupt        |
| In3     | uart_1   | External UART interrupt         |
| In4     | uart_USB | USB UART interrupt              |
| In5     | INT_0_3  | External GPIO interrupt (4-bit) |
| In6     | i2c_0    | I2C controller interrupt        |
| In7     | spi_0    | External SPI master interrupt   |

### 1.5 Debug Module ( mdm\_1 )

| Parameter  | Value                       |
|------------|-----------------------------|
| IP Version | xilinx.com:ip:mdm_riscv:1.0 |

**Description:** MicroBlaze Debug Module providing JTAG debug access for breakpoints, register inspection, and memory read/write. Its Debug\_SYS\_Rst signal is connected to the system reset module.

### 1.6 Clocking Wizard ( clk\_wiz\_1 )

| Parameter           | Value                        |
|---------------------|------------------------------|
| IP Version          | xilinx.com:ip:clk_wiz:6.0    |
| Input Clock         | 12 MHz (on-board oscillator) |
| Output Clock        | 100 MHz                      |
| PLL Multiply Factor | MMCM_CLKFBOUT_MULT_F = 62.5  |
| PLL Divide Factor   | MMCM_CLKOUT0_DIVIDE_F = 7.5  |

**Description:** Uses an MMCM/PLL to multiply the 12 MHz board clock to 100 MHz for the entire system. The Locked signal indicates clock stability.

### 1.7 Processor System Reset ( rst\_clk\_wiz\_1\_100M )

| Parameter      | Value                             |
|----------------|-----------------------------------|
| IP Version     | xilinx.com:ip:proc_sys_reset:5.0  |
| External Reset | On-board push button (Active Low) |

**Description:** Generates synchronized reset signals ( mb\_reset , bus\_struct\_reset , peripheral\_aresetn ) ensuring all modules are released from reset only after the clock is stable.

---

## 2. UART Communication

### 2.1 USB UART ( `uart_USB` )

| Parameter        | Value                                        |
|------------------|----------------------------------------------|
| IP Version       | <code>xilinx.com:ip:axi_uart16550:2.0</code> |
| AXI Base Address | <code>0x4030_0000</code>                     |
| Address Range    | 64 KB (0x4030_0000 – 0x4030_FFFF)            |
| TX Pin           | J18 (via Micro-USB connector)                |
| RX Pin           | J17 (via Micro-USB connector)                |
| Interrupt        | Connected to <code>xlconcat In4</code>       |

**Description:** 16550-compatible UART for host PC communication through the on-board Micro-USB connector. Baud rate is software-configured via the Divisor Latch Register. At 100 MHz system clock, the divisor for 115200 baud is 54 (0x36). Typically mapped as STDIN/STDOUT.

### 2.2 External UART ( `uart_1` )

| Parameter        | Value                                        |
|------------------|----------------------------------------------|
| IP Version       | <code>xilinx.com:ip:axi_uart16550:2.0</code> |
| AXI Base Address | <code>0x4031_0000</code>                     |
| Address Range    | 64 KB (0x4031_0000 – 0x4031_FFFF)            |
| TX Pin           | J1 (DIP Pin 11)                              |
| RX Pin           | K2 (DIP Pin 12)                              |
| Interrupt        | Connected to <code>xlconcat In3</code>       |

**Description:** Second 16550 UART exposed on the DIP connector for communication with external devices (e.g., sensor modules, Bluetooth modules).

---

## 3. GPIO (General Purpose I/O)

All GPIO instances use `xilinx.com:ip:axi_gpio:2.0`.

### 3.1 On-Board GPIO

| Instance                     | AXI Base Address         | Width | Direction | Connection | Description       |
|------------------------------|--------------------------|-------|-----------|------------|-------------------|
| <code>board_led_2bits</code> | <code>0x4000_0000</code> | 2     | Output    | A17, C16   | On-board LEDs × 2 |

---

| Instance                  | AXI Base Address         | Width | Direction | Connection    | Description              |
|---------------------------|--------------------------|-------|-----------|---------------|--------------------------|
| <code>board_button</code> | <code>0x4001_0000</code> | 1     | Input     | A18           | On-board push button × 1 |
| <code>board_rgb</code>    | <code>0x4002_0000</code> | 3     | Output    | B17, B16, C17 | On-board RGB LED (R/G/B) |

### 3.2 DIP Connector GPIO (4 Groups × 7-bit)

| Instance                | AXI Base Address         | Width | DIP Pins  | Description                     |
|-------------------------|--------------------------|-------|-----------|---------------------------------|
| <code>gpio_A_0_6</code> | <code>0x4003_0000</code> | 7     | Pin 1–7   | GPIO Group A, bidirectional I/O |
| <code>gpio_B_0_6</code> | <code>0x4004_0000</code> | 7     | Pin 17–23 | GPIO Group B, bidirectional I/O |
| <code>gpio_C_0_6</code> | <code>0x4005_0000</code> | 7     | Pin 42–48 | GPIO Group C, bidirectional I/O |
| <code>gpio_D_0_6</code> | <code>0x4006_0000</code> | 7     | Pin 26–32 | GPIO Group D, bidirectional I/O |

**Description:** Each group is a 7-bit bidirectional port ( `C_ALL_OUTPUTS = 0` ). Pin direction is set through the TRI register; data is read/written via the DATA register.

### 3.3 External Interrupt Inputs ( `INT_0_3` )

| Parameter            | Value                                                            |
|----------------------|------------------------------------------------------------------|
| AXI Base Address     | <code>0x4007_0000</code>                                         |
| Width                | 4-bit (all inputs)                                               |
| Interrupt Capability | Enabled ( <code>C_INTERRUPT_PRESENT = 1</code> )                 |
| DIP Pins             | Pin 8 (INTR_0), Pin 9 (INTR_1), Pin 41 (INTR_2), Pin 33 (INTR_3) |

**Description:** Four external interrupt inputs grouped into a single AXI GPIO instance. The interrupt signal is routed to `xlconcat In5` and then to the AXI Interrupt Controller.

## 4. Timers & PWM

All timer instances use `xilinx.com:ip:axi_timer:2.0`.

### 4.1 System Timers

| Instance             | AXI Base Address         | Mode               | Interrupt                 | Description                  |
|----------------------|--------------------------|--------------------|---------------------------|------------------------------|
| <code>timer_0</code> | <code>0x4010_0000</code> | 32-bit ( Default ) | <code>xlconcat In0</code> | General-purpose system timer |

| Instance | AXI Base Address | Mode    | Interrupt       | Description           |
|----------|------------------|---------|-----------------|-----------------------|
| timer_1  | 0x4011_0000      | Default | xlconcat<br>In1 | General-purpose timer |
| timer_2  | 0x4012_0000      | Default | xlconcat<br>In2 | General-purpose timer |

## 4.2 PWM Outputs

| Instance | AXI Base Address | Output Pin | DIP Pin | Description   |
|----------|------------------|------------|---------|---------------|
| PWM_0    | 0x4020_0000      | J3         | Pin 10  | PWM Channel 0 |
| PWM_1    | 0x4021_0000      | W3         | Pin 34  | PWM Channel 1 |
| PWM_2    | 0x4022_0000      | W4         | Pin 40  | PWM Channel 2 |

**Description:** AXI Timer instances configured in PWM mode to generate square-wave outputs. Useful for LED dimming, motor speed control, buzzer tone generation, etc. Frequency and duty cycle are configured through the Timer Load Registers and the PWM enable bit.

## 5. External Memory

### 5.1 QSPI Flash ( axi\_quad\_spi\_0 )

| Parameter               | Value                                                                 |
|-------------------------|-----------------------------------------------------------------------|
| IP Version              | xilinx.com:ip:axi_quad_spi:3.2                                        |
| AXI_LITE Base (Control) | 0x4050_0000 (64 KB)                                                   |
| Mode                    | Standard SPI controller ( C_XIP_MODE = 0 ) — <b>not</b> memory-mapped |
| SPI Memory Type         | Macronix ( C_SPI_MEMORY = 4 )                                         |
| Interface               | Quad SPI, FIFO depth 256                                              |

**Description:** Controls the on-board Quad-SPI NOR Flash. The controller runs in **standard (register) mode**: the full register set (control, status, TX/RX FIFO, slave select — see PG153) is exposed at 0x4050\_0000 via microblaze\_riscv\_0\_axi\_periph M19, so the CPU can issue any SPI command — read ( 0x0B ), Write Enable ( 0x06 ), Sector/Block Erase ( 0x20 / 0xD8 ), Page Program ( 0x02 ), status poll ( 0x05 ). This is what allows the UART bootloader to program application images into flash at runtime ( workspace-example/boot loader/src/boot loader .c is a complete worked example, and the Vitis xSpi driver wraps the register protocol).

Flash contents are **not memory-mapped** in this mode — there is no XIP window, and code cannot execute from flash directly. The standalone-boot design instead copies the application from flash into SRAM at

power-on (see the Standalone Boot Mode guide). The 256-entry FIFO matches the flash's 256-byte page size, so one page program fits in a single transfer.

**Design note:** XIP mode ( `C_XIP_MODE = 1` , a read-only memory-mapped window) and CPU-programmable register mode are **mutually exclusive** in this IP (PG153). This project chooses register mode: self-programming (Arduino-style `upload.py` workflow) was judged more valuable for the course than execute-in-place, and the 512 KB SRAM + I-cache serves execution instead.

5.2 SRAM / Cellular RAM ( `axi_emc_0` )

| Parameter         | Value                                         |
|-------------------|-----------------------------------------------|
| IP Version        | <code>xilinx.com:ip:axi_emc:3.0</code>        |
| AXI Base Address  | <code>0x6000_0000</code>                      |
| Address Range     | 512 KB, exact fit (0x6000_0000 – 0x6007_FFFF) |
| Physical Capacity | 512 KB Cellular RAM                           |

**Description:** External memory controller providing access to the on-board 512 KB SRAM. Suitable for large data buffers, but access latency is higher than Block RAM.

6. Analog-to-Digital Conversion (XADC)

6.1 XADC Wizard ( `xadc_wiz_0` )

| Parameter        | Value                                   |
|------------------|-----------------------------------------|
| IP Version       | <code>xilinx.com:ip:xadc_wiz:3.3</code> |
| AXI Base Address | <code>0x4060_0000</code>                |
| Address Range    | 64 KB                                   |
| Conversion Rate  | 500 KSPS                                |
| Sequencer Mode   | Continuous                              |

Enabled Channels:

| Channel     | Source             | Description                                                 |
|-------------|--------------------|-------------------------------------------------------------|
| VAUX4       | DIP Pin 15 (G3/G2) | External analog input 0 (on-board divider: 0–3.3 V → 0–1 V) |
| VAUX12      | DIP Pin 16 (H2/J2) | External analog input 1 (on-board divider: 0–3.3 V → 0–1 V) |
| Temperature | Internal           | FPGA die temperature monitor                                |
| VCCINT      | Internal           | Core voltage monitor (1.0 V)                                |
| VCCAUX      | Internal           | Auxiliary voltage monitor (1.8 V)                           |

**Description:** The Artix-7 built-in 12-bit ADC capable of measuring external analog signals and monitoring internal FPGA temperature and supply voltages. External input pins pass through an on-board resistive voltage divider (2.32 K $\Omega$  / 1 K $\Omega$ , ratio  $\approx$  0.301), accepting up to 3.3 V at the DIP pin.

**Effective Per-Channel Sampling Rate:** The sequencer continuously cycles through all 5 enabled channels (VAUX4, VAUX12, Temperature, VCCINT, VCCAUX). The aggregate conversion rate is 500 KSPS, giving each channel an effective rate of  $500\text{ K} \div 5 = \mathbf{100\text{ KSPS}}$ .

## 7. Serial Expansion Interfaces (I2C / SPI)

### 7.1 I2C Controller ( `i2c_0` )

| Parameter        | Value                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------|
| IP Version       | <code>xilinx.com:ip:axi_iic:2.1</code>                                                                                     |
| AXI Base Address | <code>0x4070_0000</code>                                                                                                   |
| SCL Frequency    | 100 kHz (standard mode)                                                                                                    |
| SCL / SDA Pins   | DIP Pin 13 (L1) / Pin 14 (L2)                                                                                              |
| Pull-ups         | Weak FPGA internal pull-ups enabled in XDC; <b>external 4.7 k<math>\Omega</math> to 3.3 V recommended</b> for real devices |
| Interrupt        | Connected to <code>xlconcat In6</code>                                                                                     |

**Description:** AXI IIC master for external I2C devices (sensors, EEPROMs, OLED displays). Open-drain signaling: any device may only pull the line low; the pull-up resistor returns it high. Devices are addressed by their 7-bit I2C address. Use the Vitis `XIic` driver.

### 7.2 External SPI Master ( `spi_0` )

| Parameter        | Value                                                                               |
|------------------|-------------------------------------------------------------------------------------|
| IP Version       | <code>xilinx.com:ip:axi_quad_spi:3.2</code> (standard mode, no STARTUP)             |
| AXI Base Address | <code>0x4080_0000</code>                                                            |
| SCLK             | 100 MHz / 16 = <b>6.25 MHz</b> (fixed at synthesis, <code>C_SCK_RATIO = 16</code> ) |
| Slave Selects    | 2 ( <code>C_NUM_SS_BITS = 2</code> ) — two devices can share the bus                |
| Pins             | DIP 35 SCLK (V3) · 36 MOSI (W5) · 37 MISO (V4) · 38 SS0 (U4) · 39 SS1 (V5)          |
| Interrupt        | Connected to <code>xlconcat In7</code>                                              |

**Description:** SPI master for external devices (displays, ADCs, flash modules). Full-duplex push-pull signaling — no pull-ups needed; the active device is chosen by driving its SS line low. Use the Vitis `xspi` driver.



**Note:** This is a *second, independent* SPI controller — do not confuse it with `axi_quad_spi_0` ( `0x4050_0000` ), which is dedicated to the on-board QSPI boot flash. Firmware should select controllers by instance macro ( `XPAR_SPI_0_BASEADDR` vs `XPAR_AXI_QUAD_SPI_0_BASEADDR` ), never by generic `XPAR_XSPI_n_*` numbering.

## 8. Complete Address Map

Peripheral addresses follow a class-based convention — `0x40[C]x_xxxx` , where **C** is the peripheral class (1 MB per class, 64 KB per instance). Reading an address immediately tells you what kind of device it is: class 0 = GPIO, 1 = Timer, 2 = PWM, 3 = UART, 4 = INTC, 5 = QSPI control, 6 = XADC, 7 = I2C, 8 = SPI.

| AXI Base Address         | Range       | Peripheral                                                               | IP Type                    | Category      |
|--------------------------|-------------|--------------------------------------------------------------------------|----------------------------|---------------|
| <code>0x0000_0000</code> | 128K / 128K | Local Memory (BRAM)                                                      | <code>blk_mem_gen</code>   | Memory        |
| <code>0x4000_0000</code> | 64K         | <code>board_led_2bits</code>                                             | <code>axi_gpio</code>      | GPIO          |
| <code>0x4001_0000</code> | 64K         | <code>board_button</code>                                                | <code>axi_gpio</code>      | GPIO          |
| <code>0x4002_0000</code> | 64K         | <code>board_rgb</code>                                                   | <code>axi_gpio</code>      | GPIO          |
| <code>0x4003_0000</code> | 64K         | <code>gpio_A_0_6</code>                                                  | <code>axi_gpio</code>      | GPIO          |
| <code>0x4004_0000</code> | 64K         | <code>gpio_B_0_6</code>                                                  | <code>axi_gpio</code>      | GPIO          |
| <code>0x4005_0000</code> | 64K         | <code>gpio_C_0_6</code>                                                  | <code>axi_gpio</code>      | GPIO          |
| <code>0x4006_0000</code> | 64K         | <code>gpio_D_0_6</code>                                                  | <code>axi_gpio</code>      | GPIO          |
| <code>0x4007_0000</code> | 64K         | <code>INT_0_3</code>                                                     | <code>axi_gpio</code>      | Interrupt     |
| <code>0x4010_0000</code> | 64K         | <code>timer_0</code>                                                     | <code>axi_timer</code>     | Timer         |
| <code>0x4011_0000</code> | 64K         | <code>timer_1</code>                                                     | <code>axi_timer</code>     | Timer         |
| <code>0x4012_0000</code> | 64K         | <code>timer_2</code>                                                     | <code>axi_timer</code>     | Timer         |
| <code>0x4020_0000</code> | 64K         | <code>PWM_0</code>                                                       | <code>axi_timer</code>     | PWM           |
| <code>0x4021_0000</code> | 64K         | <code>PWM_1</code>                                                       | <code>axi_timer</code>     | PWM           |
| <code>0x4022_0000</code> | 64K         | <code>PWM_2</code>                                                       | <code>axi_timer</code>     | PWM           |
| <code>0x4030_0000</code> | 64K         | <code>uart_USB</code>                                                    | <code>axi_uart16550</code> | Communication |
| <code>0x4031_0000</code> | 64K         | <code>uart_1</code>                                                      | <code>axi_uart16550</code> | Communication |
| <code>0x4040_0000</code> | 64K         | <code>axi_intc</code>                                                    | <code>axi_intc</code>      | System        |
| <code>0x4050_0000</code> | 64K         | <code>axi_quad_spi_0</code> ( <code>AXI_LITE</code> , full register set) | <code>axi_quad_spi</code>  | Memory        |
| <code>0x4060_0000</code> | 64K         | <code>xadc_wiz_0</code>                                                  | <code>xadc_wiz</code>      | ADC           |

| AXI Base Address | Range | Peripheral                                 | IP Type      | Category      |
|------------------|-------|--------------------------------------------|--------------|---------------|
| 0x4070_0000      | 64K   | i2c_0 (DIP 13/14)                          | axi_iic      | Communication |
| 0x4080_0000      | 64K   | spi_0 (external master, DIP 35–39)         | axi_quad_spi | Communication |
| 0x6000_0000      | 512K  | axi_emc_0 (exact physical fit, I/D-cached) | axi_emc      | Memory        |

---